KLH UNIVERSITY HYDERABAD



PROJECT PRESENTATION

Department of Computer Science and Engineering

Machine Learning · ML Sec-3

Team Number
T.N. - 24

Guide Name
Mr. Shaik Asif

Project Title : Customer Purchase Prediction & Personalized Product Recommendation

INTRODUCTION

- **Overview:** ML system that predicts if a visitor will buy a product and ranks products by that probability.
- **Problem:** thousands of products; generic best-sellers ignore a visitor's current intent.
- **Objectives:** leak-free purchase target, behavioural features, model-driven Top-N recommender, live demo.
- **Scope:** 885,129 real events from an electronics store (Sep 2020 – Feb 2021).

RESEARCH GAP

- Collaborative filtering and matrix factorisation rely on long-run user–item history.
- Unexplored: combining purchase likelihood with recommendation.
- Need: recommendations that react to near-term purchase intent.
- **Our work:** rank real products by a trained model's purchase probability.

CURRENT RESEARCH → OUR SOLUTION

ALGORITHM

Start

Input Data
events.csv (view, cart, purchase)

Preprocessing
de-duplicate, clean, keep "unknown" category / brand

Feature Engineering
33 behaviour features from past events only

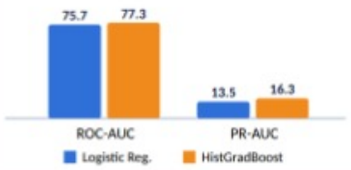
HistGradientBoosting
7-day purchase target, time-based split; Logistic Reg. baseline

Rank & Recommend
Top-N by purchase probability

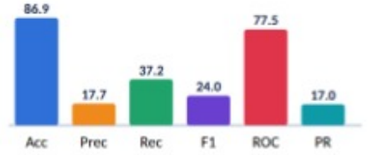
End

GRAPHS

Model Comparison (validation)



Test-Set Metrics (%)



Confusion Matrix (test)

	Pred: no	Pred: buy
Actual no	TN 76,558	FP 8,644
Actual buy	FN 3,149	TP 1,861

Event Funnel (% of events)



Innovation | Research | Impact

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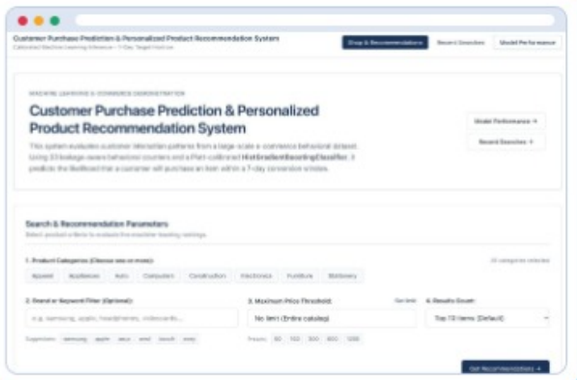


RESULTS – FRONT END

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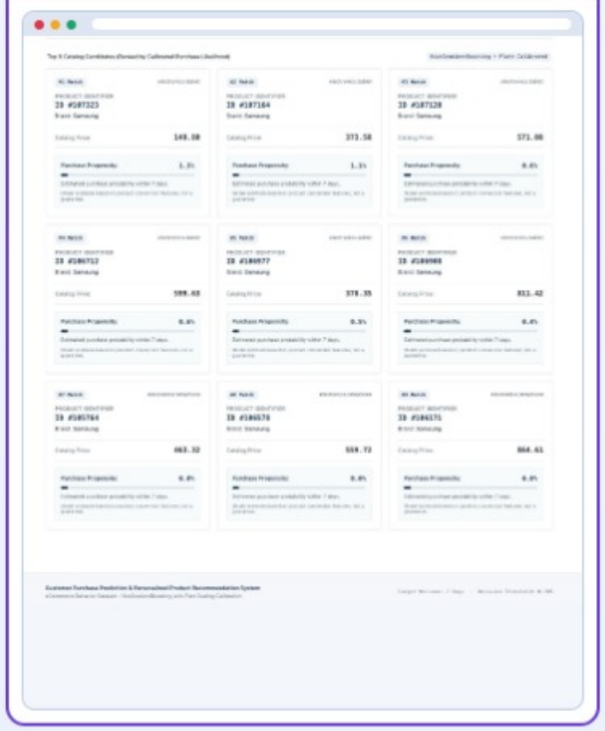
1. Home Page



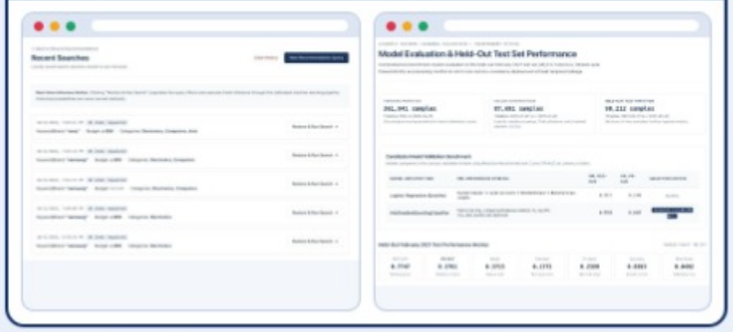
2. Filter Selection



3. Recommendations



4. Additional Screens



Key Front End Features

- Multi-select category filters
- Brand / keyword suggestions
- Price limit and Top-N options
- Ranking by calibrated 7-day probability
- Propensity bar on each product
- Recent searches saved & re-runnable

Clean interface for real-time ML recommendations.

Innovation | Research | Impact